

Semantic Theory 2026: Exercise 6 Key

Question 1

A model structure for plural terms is a tuple $M = ((U_M, \leq), V_M)$, where $(U_M, \leq)$ is an atomic join semi-lattice with universe U_M and individual part relation $\leq$, and V_M is an interpretation function mapping elements of the logical language to elements of the universe.

Consider the model M_1 , where the universe U_{M_1} is generated by the following set of atoms: $\{a, b, j, m, s\}$.

- a. Assume that $\llbracket \text{John, Mary, and Bill sing} \rrbracket^{M_1} = 1$, $\llbracket \text{Albert sings} \rrbracket^{M_1} = 1$, and $\llbracket X \text{ sing}(s) \rrbracket^{M_1} = 0$ for all other individuals (and proper sums) X for the predicate $\text{sing} \in P_d$. Then:

$$\begin{aligned} V_{M_1}(\text{sing}) &= \text{the join semi-lattice generated by } \{j, m, b, a\} \\ &= \{j, m, b, a, j \sqcup m, j \sqcup b, j \sqcup a, m \sqcup b, m \sqcup a, b \sqcup a, j \sqcup m \sqcup b, j \sqcup m \sqcup a, j \sqcup b \sqcup a, m \sqcup b \sqcup a, j \sqcup m \sqcup b \sqcup a\} \end{aligned}$$

- b. Assume that $\llbracket \text{John and Mary meet} \rrbracket^{M_1} = 1$, $\llbracket \text{Albert and Sally meet} \rrbracket^{M_1} = 1$, $\llbracket \text{Bill and Mary meet} \rrbracket^{M_1} = 1$, and $\llbracket X \text{ meet} \rrbracket^{M_1} = 0$ for all other individuals (and proper sums) X for the predicate $\text{meet} \in P_c$. Then:

$$V_{M_1}(\text{meet}) = \{j \sqcup m, a \sqcup s, b \sqcup m\}$$

Question 2

Consider the following sentence: $S = \text{“two students summarized three papers”}$

- a. How many readings does the sentence S have? List all possible readings in natural language.

Let:

- D_s = the distributive reading of “two students”
- C_s = the collective reading of “two students”
- D_p = the distributive reading of “three papers”
- C_p = the collective reading of “three papers”

We have the following readings:

- i. $C_s > C_p$: two students together wrote one summary that summarizes all three papers
 - ii. $C_s > D_p$: two students together wrote one summary for each of the three papers
 - iii. $D_s > C_p$: two students each wrote one summary that summarizes all three papers (not necessarily the same three for both students)
 - iv. $D_s > D_p$: two students each wrote one summary for each of the three papers (not necessarily the same three for both students)
 - v. $C_p > C_s$: same as (i)
 - vi. $C_p > D_s$: two students each wrote one summary that summarizes all three papers (the same three for both students)
 - vii. $D_p > C_s$: for each of the three papers, there was a group of two students that summarized that paper together
 - viii. $D_p > D_s$: for each of the three papers, there were two students who individually summarized that paper
- b. Translate each reading of S to the extended first-order logic for plural terms introduced in the lecture, which extends first-order logic with a $\oplus$ operator, a $\triangleleft$ operator, and variables $X, Y, Z, \dots$ ranging over proper sums: $X \oplus Y$ denotes the group consisting of X and Y , $\triangleleft$ denotes the part-of-relation.

You may (and should) also incorporate the function $N(X) = |\{y \mid At(y) \wedge y \triangleleft X\}|$ that takes a proper sum X and returns the number of atoms in X .

- i. $(C_s > C_p)$:
 $\exists X \exists Y [student_{PL}(X) \wedge N(X) = 2 \wedge paper_{PL}(Y) \wedge N(Y) = 3 \wedge summarize(X, Y)]$
- ii. $(C_s > D_p)$:
 $\exists X \exists Y [student_{PL}(X) \wedge N(X) = 2 \wedge paper_{PL}(Y) \wedge N(Y) = 3 \wedge \forall w [(At(w) \wedge w \triangleleft Y) \rightarrow summarize(X, w)]]$
- iii. $(D_s > C_p)$:
 $\exists X [student_{PL}(X) \wedge N(X) = 2 \wedge \forall z [(At(z) \wedge z \triangleleft X) \rightarrow \exists Y [paper_{PL}(Y) \wedge N(Y) = 3 \wedge summarize(z, Y)]]]$
- iv. $(D_s > D_p)$:
 $\exists X [student_{PL}(X) \wedge N(X) = 2 \wedge \forall z [(At(z) \wedge z \triangleleft X) \rightarrow \exists Y [paper_{PL}(Y) \wedge N(Y) = 3 \wedge \forall w [(At(w) \wedge w \triangleleft Y) \rightarrow summarize(z, w)]]]]]$
- v. $(C_p > C_s)$: same as (i)
- vi. $(C_p > D_s)$:
 $\exists Y [paper_{PL}(Y) \wedge N(Y) = 3 \wedge \exists X [student_{PL}(X) \wedge N(X) = 2 \wedge \forall z [(At(z) \wedge z \triangleleft X) \rightarrow summarize(z, Y)]]]$
- vii. $(D_p > C_s)$:
 $\exists Y [paper_{PL}(Y) \wedge N(Y) = 3 \wedge \forall w [(At(w) \wedge w \triangleleft Y) \rightarrow \exists X [student_{PL}(X) \wedge N(X) = 2 \wedge summarize(X, w)]]]$
- viii. $(D_p > D_s)$:
 $\exists Y [paper_{PL}(Y) \wedge N(Y) = 3 \wedge \forall w [(At(w) \wedge w \triangleleft Y) \rightarrow \exists X [student_{PL}(X) \wedge N(X) = 2 \wedge \forall z [(At(z) \wedge z \triangleleft X) \rightarrow summarize(z, w)]]]]]$